Radau: Numerical Method and Adaptive Execution

1 Differential equation and numerical method

Consider $z' = f(t, z)$, $z(t_0) = z_0$, in the project's packed physical layout. A three-stage Radau IIA collocation method of order five, with an embedded third-order error estimator and a cubic collocation interpolant. The implementation delegates the numerical tableau, local error estimator, trial rejection and interpolation polynomial to the installed SciPy `Radau` solver. The adapter does not duplicate these algorithms.

A Runge–Kutta evaluation has stage equations

$$Y_i = z_n + h \sum_j a_{ij} f(t_n + c_j h, Y_j), \qquad z_{n+1} = z_n + h \sum_i b_i f(t_n + c_i h, Y_i).$$

The collocation equations are implicit and are solved by SciPy Newton iterations. SciPy accepts an optional physical-state Jacobian callable; otherwise SciPy uses finite differences. Its dimension is the physical state dimension.

2 Error control and accepted intervals

The adaptive backend scales local error estimates using positive absolute and relative tolerances, with component scale based on $\mathrm{atol} + \mathrm{rtol} \max(|z_n|, |z_{n+1}|)$. It selects subsequent step sizes and can reject trial steps. `max_step` bounds accepted duration; `first_step=None` delegates initial selection to the backend. Requested tolerances bound neither global trajectory error nor roundoff.

One live solver is retained over the complete interval. Its state, stage work and any Jacobian/factorization reuse survive accepted steps. The adapter calls `solver.step()` until acceptance or failure; rejected trials never become observer records or separate accepted metric rows.

3 Standalone reference calculations

This method is available through SciPy directly in reference studies, not through GC2D's IntegrationMethod or simulation facade. The reference pipeline records physical trajectories, requested dense samples and SciPy work counters. The recurrence study retains a dense solution for return searches. There is no adaptive GC2D controller or SciPy/JAX hybrid execution path.

Historical generalized-energy comparisons may separately apply an eight-node Gauss rule to accepted dense interpolants to reconstruct $\dot{\kappa} = -\partial_t H$. This diagnostic has no independent adaptive error tolerance; refinement must audit its accuracy. Existing archives retain their original interpretation. Generic fixed-step EnergyObserver instead records physical H and positive analytic $\partial_t H$ at endpoints and saved samples.